Name _____ Period _____ Date _____

2-7 Lesson Quiz: Linear-Quadratic Systems

short form

This form has Savvas items 2, 3 and 4. Items 1 and 5 are on the full Savvas quiz only.

2. Use substitution to solve the system $y = -\frac{1}{2}x^2$, $y = x - 4$.

Ⓐ -4 and 2

Ⓑ $(4, -8)$ and $(-2, -2)$

Ⓒ $(-4, -8)$ and $(2, -2)$

Ⓓ $(0, 0)$ and $(0, -4)$

Show your substitution:

_____

_____

_____

3. Nate tosses a ball up a hill for his dog to chase. The path of the ball is modeled by the function $y = -\frac{1}{4}x^2 + \frac{33}{5}x$, where x is the ball's horizontal distance from Nate in feet and y is the ball's height in feet. The hill is modeled by the line $y = \frac{1}{5}x$. How far does the ball travel horizontally before it hits the ground?

Show your work:

_____

_____

_____

The ball travels about

_____ ft horizontally.

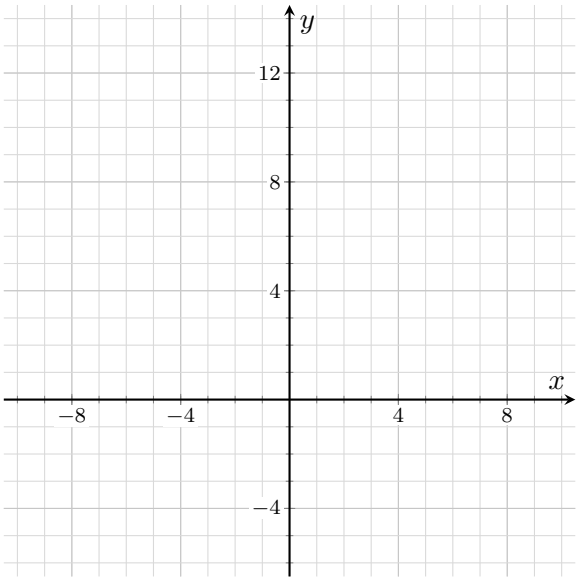
4. Solve the system of inequalities $y \geq 2x^2 + 2x - 3$, $y \leq \frac{1}{3}x + 6$ using shading.

Optional check (not scored): pick a point in your shaded region and show it makes both inequalities true.

_____

_____

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Scoring: Questions 3 and 4 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

2-7 Lesson Quiz, short form: answer key

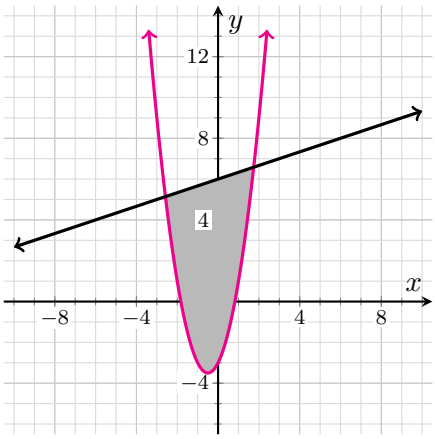
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2. C, $(-4,-8)$ and $(2,-2)$. $-\frac{1}{2}x^2 = x - 4$, so $x^2 + 2x - 8 = 0$, $(x+4)(x-2) = 0$, $x = -4$ or $x = 2$; then $y = x - 4$ gives -8 and -2. Distractor A is the x-values without their y-values; B swaps the signs of x. DOK 2, Realize skills A91, A54. **Bonus 5:** 5 for C with the substitution shown; 3 for C alone; 2 for A with correct work (the x-values found, not paired).

3. 25.6 ft. $-\frac{1}{4}x^2 + \frac{33}{5}x = \frac{1}{5}x$, so $-\frac{1}{4}x^2 + \frac{32}{5}x = 0$, $x\left(-\frac{1}{4}x + \frac{32}{5}\right) = 0$, $x = 0$ or $x = \frac{128}{5} = 25.6$. $x = 0$ is the toss; the ball lands on the hill at $x = 25.6$ (height 5.12 ft). The TE wording says "hits the ground"; the ground is the hill. DOK 1, Realize skills A91, A54. **10 points:** 3 for setting the path equal to the hill, 4 for solving to $x = 0$ and $x = 25.6$, 3 for choosing 25.6 ft with a reason ($x = 0$ is where Nate tosses it).

4. Shade above the parabola and below the line, where they overlap.

Both boundaries are **solid** ($\geq$, $\leq$). Parabola $y = 2x^2 + 2x - 3$: vertex $(-0.5, -3.5)$, y-intercept -3; shade above (inside). Line $y = \frac{1}{3}x + 6$: y-intercept 6, slope $\frac{1}{3}$; shade below. The overlap lies between the crossings, where $6x^2 + 5x - 27 = 0$: $x = \frac{-5 \pm \sqrt{673}}{12}$, about $(-2.58, 5.14)$ and $(1.75, 6.58)$. Check $(0,0)$: $0 \geq -3$ and $0 \leq 6$. DOK 2, Realize skills A91, A54. **10 points:** 3 parabola (vertex and shape), 2 line, 2 both solid, 3 the overlap shaded (the optional point check is not scored).



Items left off this form (full Savvas quiz only; available for a retake or Form B): Savvas item 1, number of real solutions of $y = x^2 - 3$, $y = -2x + 4$ (answer A, 2; $x^2 + 2x - 7 = 0$ has discriminant $32 > 0$; DOK 2, A91, A54; taught by Example 1, cut in the fast packet); Savvas item 5, solve $2x^2 + 5x - 8 = \frac{5}{2}x + 20$ with the graphing calculator's intersection feature, nearest tenth (answer $x \approx 3.2$ and $x \approx -4.4$; DOK 2, A91, A54; taught by Example 5, cut).

IXL replacement: SmartScore 70 or above on HVZ, VQU and U5D replaces this quiz with 20/20.